AUTOSAR HIGH-LEVEL DESIGN SPECIFICATION

Subsystem: Advanced Electronic Brake Control System (EBCS)

Document Version: v2.0.0

Classification: Confidential - Automotive Engineering / Embedded Systems

Target Platform: AUTOSAR Classic Platform Release 4.4.0

1. Document Overview and Scope

This High-Level Design (HLD) specification defines the software architecture, Application Software Components (SWCs), ports, interfaces, signals, and functional data flows for the Electronic Brake Control System (EBCS). The system coordinates anti-lock braking (ABS), electronic stability control (ESC), and regenerative brake blending. All interfaces conform to AUTOSAR standard methodologies and ISO 26262 ASIL-D safety requirements.

2. Architecture Baseline
The software architecture decomposes into primary functional clusters: Sensor Ingestion SWCs, Algorithmic Computation SWCs, and Actuator Driver SWCs. Communication between components occurs via Virtual Functional Bus (VFB) Sender-Receiver and Client-Server interfaces.

3. Software Component Architecture Overview

The EBCS architecture comprises the following key Software Components (SWCs):
- SpeedProcessingSWC: Ingests raw wheel speed pulse trains and computes filtered vehicle velocity.
- BrakePedalSensorSWC: Measures driver brake pedal displacement and deceleration request force.
- BrakeControlSWC: Calculates target brake pressure per wheel and traction slip limits.
- HydraulicActuatorSWC: Directly drives electro-hydraulic valve coils and pump motor relays.
- DiagnosticMonitorSWC: Performs ASIL-D plausibility checks and sensor fault isolation.

4. Component Specification: SpeedProcessingSWC

Section 4.1 Component Role
SpeedProcessingSWC executes cyclically every 5ms on Task_Periodic_5ms. It samples the front-left, front-right, rear-left, and rear-right wheel pulse counters and calculates longitudinal vehicle velocity.

Section 4.2 Port and Interface Definitions
Component: SpeedProcessingSWC
- Provide-Port: P_SpeedReceiver
  Interface: VehicleSpeed_IF (SenderReceiverInterface)
  Transmitted Signal: VehicleSpeed
  Data Type: uint16 (Scaled 0.01 km/h per LSB, Range: 0 - 350.00 km/h)
  Description: VehicleSpeed is received via port P_SpeedReceiver on component SpeedProcessingSWC and published to downstream subscribers including BrakeControlSWC and EngineControlSWC.

- Require-Port: R_WheelSpeedSensor
  Interface: WheelSpeedSensor_IF (SenderReceiverInterface)
  Received Signal: RawWheelTicks
  Data Type: uint32
  Description: Raw magnetic hall sensor pulse transitions per millisecond.

5. Component Specification: BrakeControlSWC

Section 5.1 Algorithmic Control Flow
BrakeControlSWC operates cyclically every 10ms. It ingests VehicleSpeed from SpeedProcessingSWC and BrakePedalTravel from BrakePedalSensorSWC to determine primary braking pressure.

Section 5.2 Ports and Signal Mappings
- Require-Port: R_VehicleSpeedConsumer
  Interface: VehicleSpeed_IF
  Signal: VehicleSpeed
  Data Type: uint16
  Connection: Connected to SpeedProcessingSWC.P_SpeedReceiver via VFB assembly connector.

- Require-Port: R_PedalPosition
  Interface: PedalTravel_IF
  Signal: DriverBrakeForce
  Data Type: float32 (Unit: Newtons, Range: 0.0 - 1200.0 N)

- Provide-Port: P_TargetBrakePressure
  Interface: BrakePressure_IF (SenderReceiverInterface)
  Transmitted Signal: TargetCylinderPressure
  Data Type: float32 (Unit: Bar, Range: 0.0 - 250.0 Bar)

6. Actuator Driver & Diagnostic Architecture

Section 6.1 HydraulicActuatorSWC Details
- Require-Port: R_CylinderPressureInput
  Interface: BrakePressure_IF
  Signal: TargetCylinderPressure
  Data Type: float32
- Provide-Port: P_ValvePWMOutput
  Interface: ActuatorPWM_IF
  Signal: CoilDutyCycle
  Data Type: uint8 (Range: 0 - 100 %)

Section 6.2 Diagnostic & Fault Isolation
DiagnosticMonitorSWC continuously verifies whether signal plausibility bounds are satisfied. If VehicleSpeed deceleration exceeds 1.4g without corresponding TargetCylinderPressure build-up, an ISO 26262 Diagnostic Trouble Code (DTC_0x84A12) is latched into NvM (Non-Volatile Memory).